Examine why advances in science are often the result of collaboration or build on the work of others

Learning intention

- examine why advances in science are often the result of collaboration or build on the work of others.

Teach and model

- constructing a timeline to show how contributions and collaboration of scientists.
- investigating why scientists changed the phosphate levels in detergents to prevent algal blooms.
- investigating how astronauts and scientists from many different countries have collaborated in the International Space Station program.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.